ABDELRAHMAN ELNAHRAWY

Embedded Systems & Firmware Engineer | PIC • AVR • ARM • ESP32/ESP8266 | Bare-Metal • RTOS | PCB Design (Altium / CircuitMaker, KiCad, Eagle) | UART • SPI • I²C • MQTT

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PROFESSIONAL SUMMARY

Embedded Software & Hardware Engineer with hands-on experience designing and developing end-to-end embedded systems, from schematic capture and PCB layout to bare-metal and RTOS-based firmware using C and C++. Strong expertise in ARM Cortex-M, PIC, AVR, and ESP microcontrollers. Skilled in GNU Make build systems, linker scripts, memory mapping, and startup code. Experienced in hardware bring-up, debugging, motor control, sensor integration, and embedded communication protocols including UART, SPI, I²C, and MQTT/MQTTS. Focused on reliability, performance, and clean system architecture.

WORK EXPERIENCE

Embedded Software & Hardware Engineer (Part-Time)

Armtronics — Cairo, Egypt (Hybrid) **Jun 2024 – Jan 2025**

Engineered end-to-end solutions from concept to functional prototypes.

Key Contributions:

- Developed bare-metal firmware for PIC, AVR, and ESP32/ESP8266 MCUs using C/C++.
- Implemented secure MQTT/MQTTS communication for cloud monitoring and control.
- Integrated multiple sensors and designed control units using Digital I/O and PWM.
- Managed full PCB lifecycle: schematic capture, layout, bring-up, and debugging.
- Built robust device-to-server links with real-time feedback and command handling.

Technologies:

Embedded C, C++, PIC, AVR, ARM Cortex-M, ESP32/ESP8266, MQTT
PCB Design (Altium/CircuitMaker, KiCad, Eagle), UART, SPI, I²C, Hardware Debugging

Embedded Systems Engineer

Dynamic Systems Egypt — Giza, Egypt (On-site) **Jul 2023 – Oct 2023**

Contributed to R&D of turnstile gate systems for metro and urban transportation, focusing on embedded control electronics, system reliability, and hardware–mechanical integration.

Key Contributions:

- Designed embedded control systems integrating microcontrollers (PIC / AVR), sensors, motors, relays, and actuators.
- Supported hardware development: schematic review, PCB bring-up, and hardware debugging; applied SOLIDWORKS and Design-for-Manufacturing (DFM) principles.
- Gained hands-on experience with hardware–mechanical integration and industrial deployment constraints.

Technologies:

Embedded C, Bare-Metal Programming, Microcontrollers (PIC/AVR/ARM Cortex-M)
PCB Design (Eagle), Hardware Debugging, SOLIDWORKS, DFM

EDUCATION

Bachelor of Engineering – Electrical Power & Machines
Ain Shams University — Cairo, Egypt **Graduated: 2024**

PROFESSIONAL TRAINING & CERTIFICATIONS

Advanced Embedded Systems Nanodegree – Udacity – Egypt FWD / Sprints **Aug 2022 – Nov 2022**

- Embedded C & Bare-Metal Programming
- ARM Cortex-M Architecture
- RTOS Concepts and Task Scheduling
- Embedded Software Design & MISRA Guidelines
- Embedded Software & Hardware Architecture

Embedded Software & Hardware Architecture – **University of Colorado Boulder (Coursera) Dec 2022 – Feb 2023**

- MCU architecture & memory models
- Linker scripts and build systems (GNU Make)
- Low-level driver design and system layering

Agile & Scrum Training Orange Digital Center – Egypt **Sep 2022 – Oct 2022**

- Agile methodologies, Scrum framework, and team collaboration best practices

PROJECTS

EV Motor Drive Control System (FOC & V/F) **Graduation project**

Key Contributions:

- Developed real-time FOC system for 3-phase induction motor.
- Built layered firmware with FreeRTOS for ADC, PID loops, and SVPWM.
- Designed and implemented PCB & power-stage hardware.
- Validated control strategies via MATLAB/Simulink before deployment.
- Created modular HAL for portable and maintainable firmware.

Technologies:

ARM Cortex-M4 (Tiva C), FreeRTOS, Embedded C/C++, MATLAB/Simulink,
PCB Design (Altium/CircuitMaker/KiCad), AS5600 Encoder, UART, SPI, I^2C, PID Control, SVPWM

SKILLS

Embedded Systems: C/C++, Bare-Metal, FreeRTOS, ARM Cortex-M, ESP32/8266, AVR, PIC
Communication: UART, SPI, I2C, CAN, BLE, MQTT/MQTT-SN, TCP/IP
Hardware Design: PCB Design (Altium/CircuitMaker, KiCad, Eagle), Schematic Capture, Proteus
Development Tools: Git, VS code, GNU Make, Keil, STM32Cube, ESP-IDF, TivaWare, SOLIDWORKS, Agile/Scrum